



BITS Pilani
K K Birla Goa Campus

8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

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Memory Interfacing: 8086

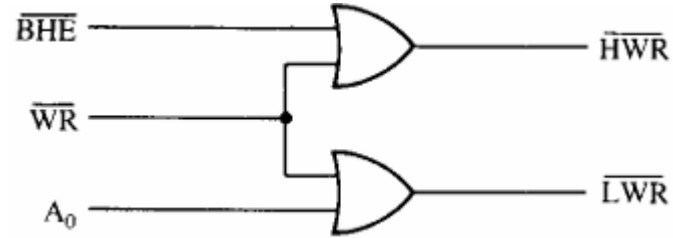
If memory chips available are only 2KB each. Do the interfacing for 8086

Ex: Interface 4K 2716 (ROM) starting at 00000H 8K 6116 (SRAM) starting at 08000

ROM₁ **00000_H - 00FFF_H**

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1

Separate Bank Write Strobes.



OR gate combines A_0 with $\overline{WR}$ for the low bank selection signal ($\overline{LWR}$)
 $\overline{WR}$ combines with $\overline{BHE}$ for the high bank selection signal ($\overline{HWR}$) .

Write strobes, for the 80286/80386SX, are generated by using the $\overline{MWTC}$ signal instead of $\overline{WR}$.

Separate Bank Write strobe

Bank selection can be effectively handled by developing separate write strobe for each memory bank using only one decoder.

Why not separate read strobe?



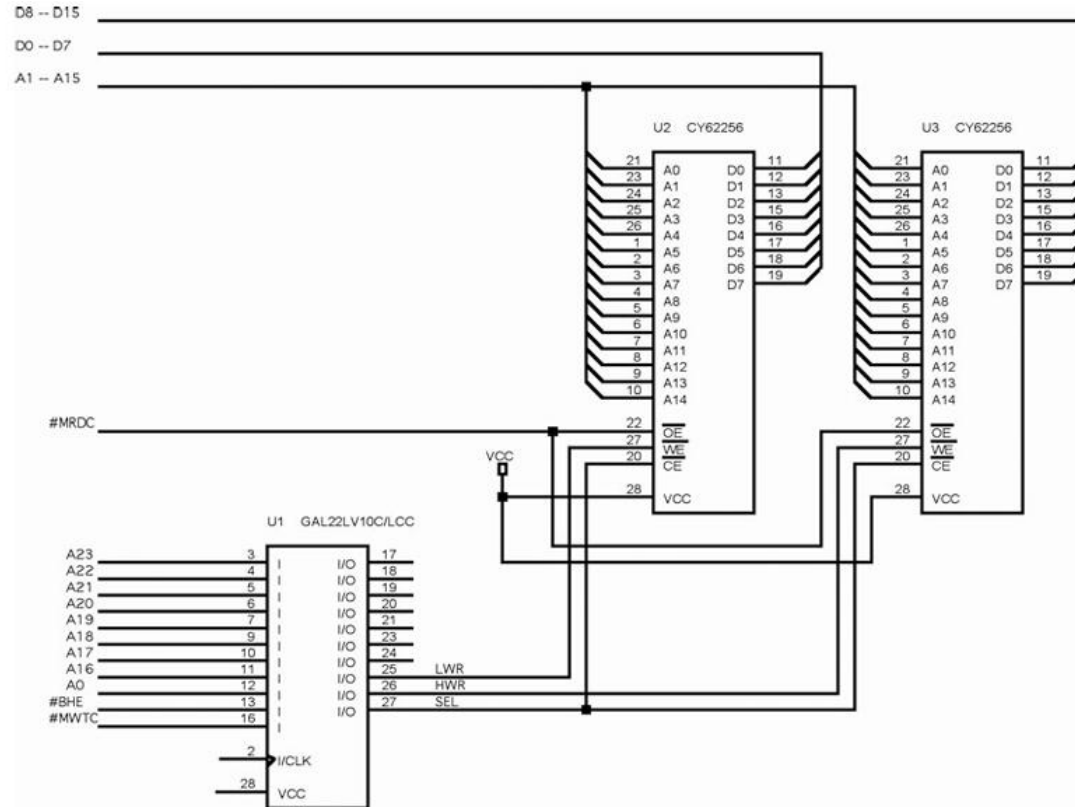
Microprocessor read only the byte of data that they need at any given time from the half of the data bus. If 16 bit sections of data are always presented to the data bus during a read the microprocessor ignores the 8 bit section that it does not need without any conflicts or problems.

Memory decoder 16bit

Decode 64K, 16-bit-wide memory stored at locations **060000H–06FFFFH** for the 80286 or 80386 microprocessor.

- For a 64K bytes SRAM requires two 32K byte (62256) memory devices Memory is decoded , so bit **A0** is a don't care for the decoder. Bit positions **A1–A15** are connected to memory component address pins **A0–A14**.
- The decoder (**GAL22V10**) enables both memory devices by using address connection **A23–A15** to select memory whenever address **06XXXXH** appears on the address bus.

16 bit Memory Decoding(GAL22V10) using separate write strobe



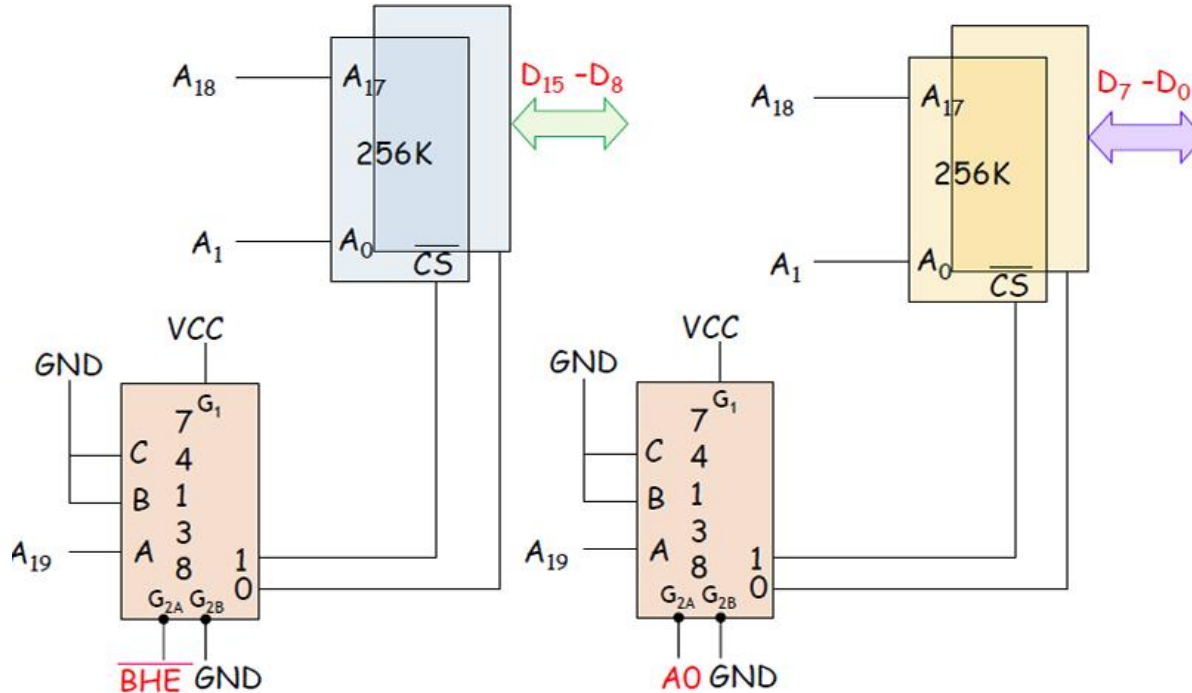
Memory Interfacing: 8086

If memory chips available are only 256KB

Ex: Interface 1MB (SRAM) to 8086

$A_{19}A_{18}A_{17}A_{16}$	$A_{15}A_{14}A_{13}A_{12}$	$A_{11}A_{10}A_9A_8$	$A_7A_6A_5A_4$	$A_3A_2A_1A_0$	
0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	} 512KB
0 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	
1 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	} 512KB
1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	

Memory Interfacing: 8086

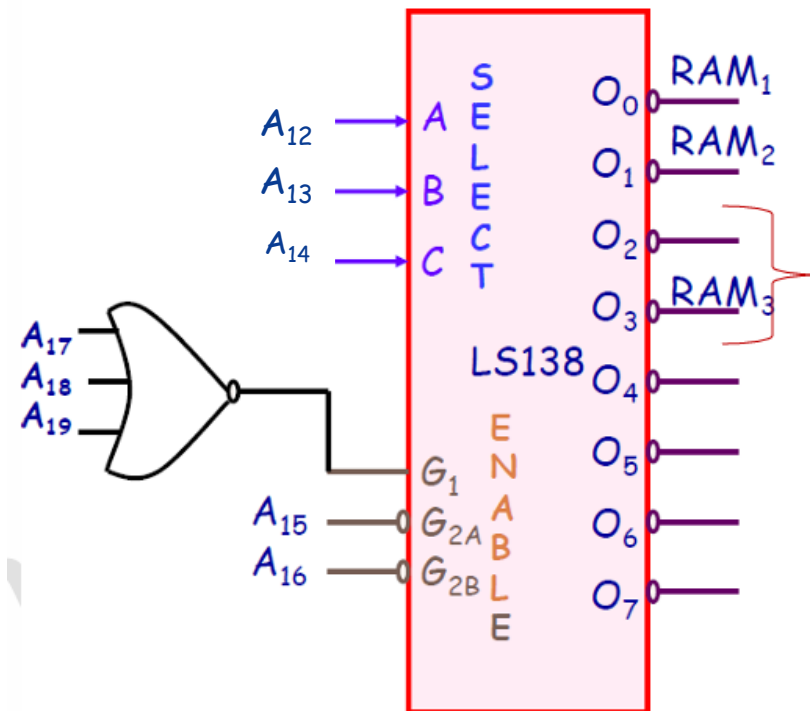


Problem: If memory chips available are 2KB (4 chips) and 4KB (2 Chips).

Ex: Interface 16KB of RAM to 8086 starting at 00000H

$RAM_{1\text{even}}$	$00000_H, 00002_H, 00004_H, \dots, 00FFE_H$
$RAM_{1\text{odd}}$	$00001_H, 00003_H, 00005_H, \dots, 00FFF_H$
$RAM_{2\text{even}}$	$01000_H, 01002_H, 01004_H, \dots, 01FFE_H$
$RAM_{2\text{odd}}$	$01001_H, 01003_H, 01005_H, \dots, 01FFF_H$
$RAM_{3\text{even}}$	$02000_H, 02002_H, 02004_H, \dots, 03FFE_H$
$RAM_{3\text{odd}}$	$02001_H, 02003_H, 02005_H, \dots, 03FFF_H$

19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1



8086 I/O Instructions

- IN and OUT transfer data between an I/O device and the microprocessor's accumulator (AL, AX or EAX).
- The I/O address is stored in: Register DX as a 16-bit I/O address (variable addressing/Indirect). The byte, p8, immediately following the opcode (fixed address/Direct).
- INS and OUTS transfer to I/O devices using ES:DI and DS:SI, respectively.
- IN AL, 19H; 8-bits data are saved to AL from I/O port 19H (Fixed address)
- IN AL, DX; 8-bits data are saved to AL from I/O port [DX] (Variable address)
- IN AX, DX; 16-bits are saved to AX.
- IN AX, 20H 16-bits data are saved to AX from I/O port 20H
- OUT DX, AX; 16-bits are written to port DX from AX
- OUT 19H, AL; 8-bits are written to I/O port 0019H.

<i>Instruction</i>	<i>Data Width</i>	<i>Function</i>
IN AL, p8	8	A byte is input into AL from port p8
IN AX, p8	16	A word is input into AX from port p8
IN EAX, p8	32	A doubleword is input into EAX from port p8
IN AL, DX	8	A byte is input into AL from the port addressed by DX
IN AX, DX	16	A word is input into AX from the port addressed by DX
IN EAX, DX	32	A doubleword is input into EAX from the port addressed by DX
INSB	8	A byte is input from the port addressed by DI and stored into the extra segment memory location addressed by DI, then $DI = DI \pm 1$
INSW	16	A word is input from the port addressed by DI and stored into the extra segment memory location addressed by DI, then $DI = DI \pm 2$
INSD	32	A doubleword is input from the port addressed by DI and stored into the extra segment memory location addressed by DI, then $DI = DI \pm 4$
OUT p8, AL	8	A byte is output from AL into port p8
OUT p8, AX	16	A word is output from AL into port p8
OUT p8, EAX	32	A doubleword is output from EAX into port p8
OUT DX, AL	8	A byte is output from AL into the port addressed by DX
OUT DX, AX	16	A word is output from AX into the port addressed by DX
OUT DX, EAX	32	A doubleword is output from EAX into the port addressed by DX
OUTSB	8	A byte is output from the data segment memory location addressed by SI into the port addressed by DX, then $SI = SI \pm 1$
OUTSW	16	A word is output from the data segment memory location addressed by SI into the port addressed by DX, then $SI = SI \pm 2$
OUTSD	32	A doubleword is output from the data segment memory location addressed by SI into the port addressed by DX, then $SI = SI \pm 4$

Direct I/O – the port address is one of the operands

– Address must be 00-FFh.

IN AL, 27h

Data flows through the accumulator

MOV AX, BX

OUT 26h, AX

; move 16-bit data from AX to port ; 26h (AL to 26h and AH to 27h)

Indirect I/O – the port address is preloaded into DX
– Address can be 0000-FFFFH

String I/O – allows data to pass directly from I/O device to memory

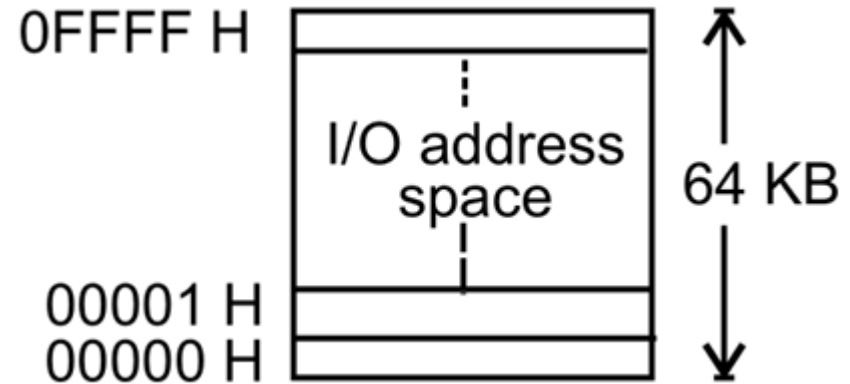
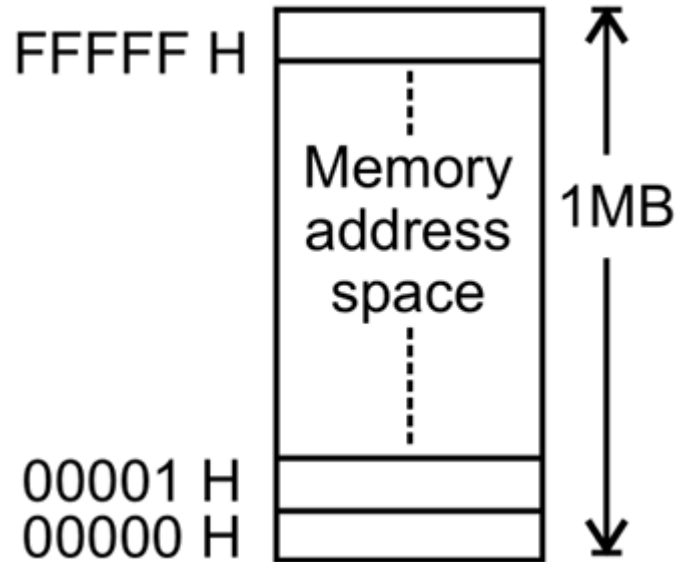
I/O Interface

I/O devices connect to processor through PORTS. IO ports are 8 bit wide so when a 16 bit port is accessed two consecutive 8-bit pots are actually addressed. Eg. If port 100H is accessed as word then 100H and 101H are actually accessed.

Ports are:

- registers (part of the I/O interface)
- 8, 16, or 32 bits wide
- Addressed in the range 0000-FFFFh
- Accessed with 2 instructions – IN, OUT

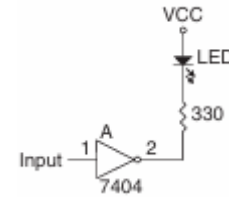
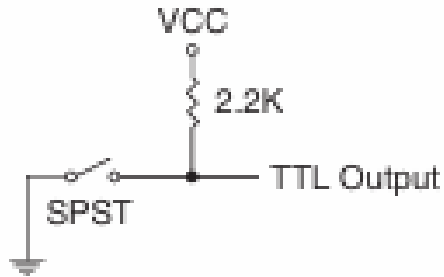
IO-Mapped & Memory-Mapped



Isolated I/O	Memory mapped I/O
1. I/O devices are treated separate from memory. 2. Full 1 MB address space is available for use as memory. 3. Separate instructions are provided in the instruction set to perform isolated I/O input-output operations. These maximise I/O operations. 4. Data transfer takes place between I/O port and AL or AX register only. This is certainly a disadvantage.	1. I/O devices are treated as part of memory. 2. Full 1 MB cannot be used as memory since I/O devices are treated as part of memory. 3. No separate instructions are needed in this case to perform memory mapped I/O operations. Hence, the advantage is that many instructions and addressing modes are available for I/O operations. 4. No such restriction in this case. Data transfer can take place between I/O port and any internal register. Here, the disadvantage is that it somewhat slows the I/O operations.

I/O Interfacing

- Input Device – Example Switch
- Output Device – LED



Input device & Output Devices interfacing

Input devices are connected with Buffer.

Why Buffers?

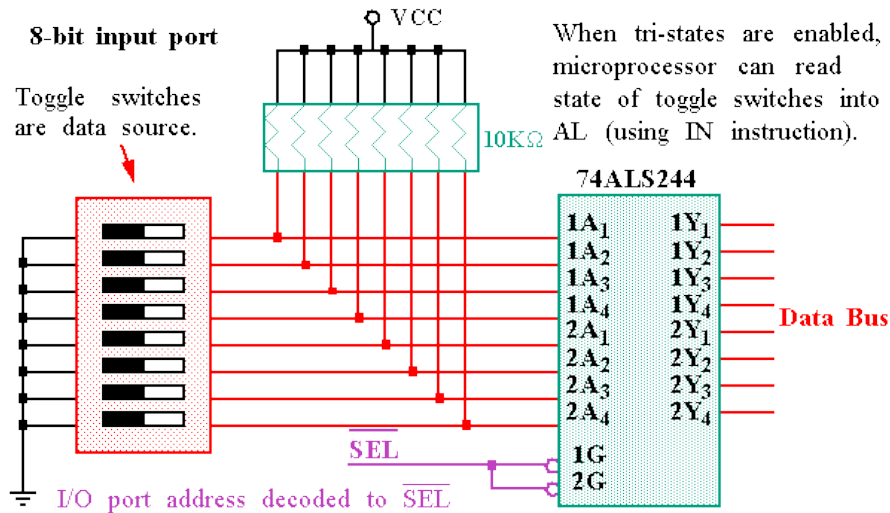
- Input Devices must be isolated from the global data bus- unwanted data may be transferred to the data bus.
- Tristate Buffers are used to provide isolation and to strengthen the signal

Output Devices are connected with Latches.

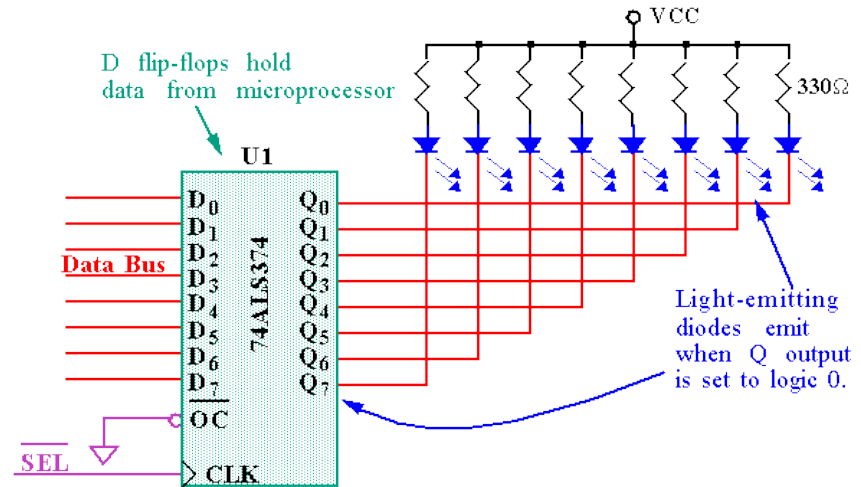
Data provided by Microprocessor is available only for short period of time (50-1000ns) data must be latched or it will be lost. Memories have internal latches.

I/O Design in 8086

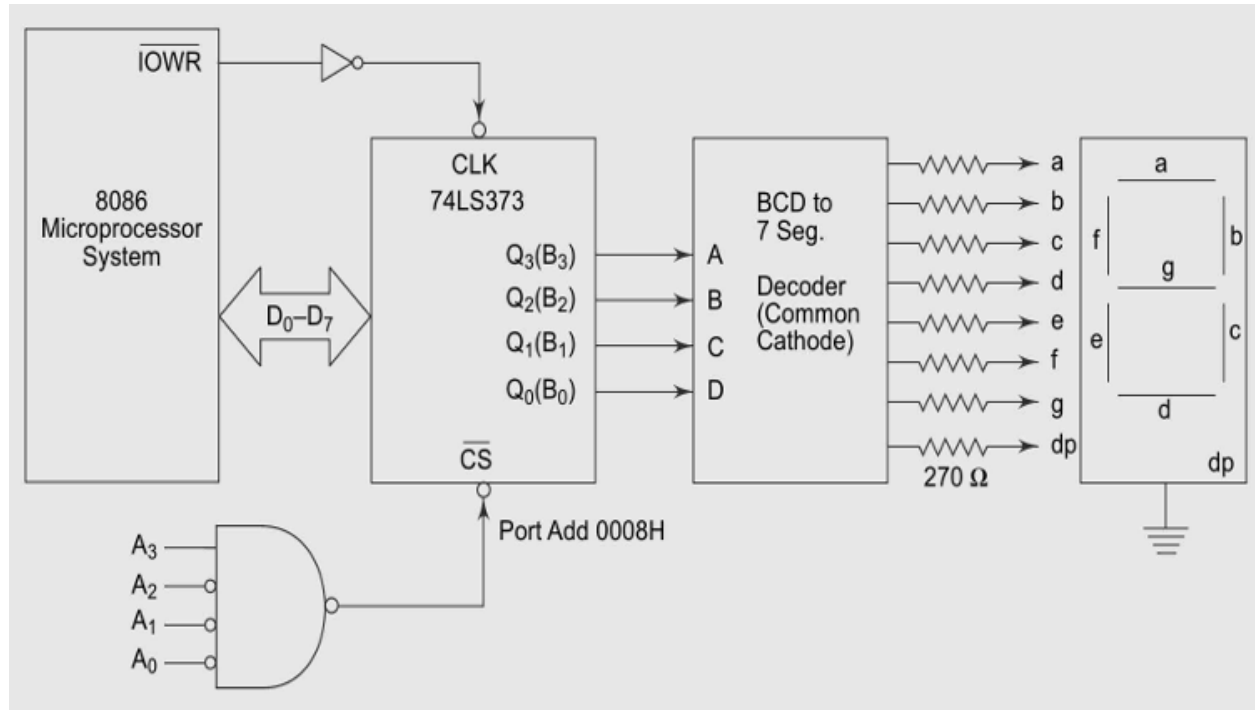
Input Interfacing



Output Interfacing



74373 : Latch



74LS245 Buffer

